Calgary Retail Fuel Price Machine Learning Project

End-to-End Workflow, Chat History Analysis, & Artifact Catalog Summary

Executive Overview

This PDF summary documents the complete machine learning modeling pipeline developed for forecasting daily retail unleaded gasoline prices in Calgary, Alberta. Spanning January 1, 2021 through October 23, 2026 (1,516 daily observations), the project establishes data cleaning protocols, feature engineering (windowing & lags), standardized temporal splitting, rigorous model comparison, AI governance safeguards, and a fully functional prediction application.

1. Comprehensive Workflow & Analysis Summary

Phase 1: Dataset Overview & Cleaning (DP1)

The raw dataset consists of 1,516 daily price logs with 0 missing values. Prices ranged from a low of **100.20 ¢/L** (Jan 2021) to an all-time peak of **191.90 ¢/L** (June 2022), averaging **142.73 ¢/L**. Crucially, exact trading days were preserved without inventing synthetic weekend/holiday data, avoiding artificial zero-change noise in autocorrelation functions.

Phase 2: Feature Engineering (DP2) & Windowing Selection

A **7-recorded-day window** (~1 trading week) was selected based on historical price dynamics:
• **Price Level Persistence:** Lag-1 autocorrelation is extremely strong (r = 0.987). Partial autocorrelation (PACF) cuts off sharply after Lag 3 (0.985 at Lag 1, -0.163 at Lag 2, +0.179 at Lag 3).
• **Momentum & Mean Reversion:** 1-day deltas show positive momentum (+0.175) reflecting station price hike spillovers, while 2-day deltas show negative correlation (-0.196) capturing 48-hour price corrections.
• **Engineered Features:** 4 direct price lags (lags 1, 2, 3, 7), 3 momentum deltas (1d, 2d, 7d diffs), and 6 rolling window metrics (7d & 14d mean, std, min, max).

Phase 3: Data Transformation & Forward Temporal Split (DP3)

Features were standardized using Z-score scaling strictly fitted on training data (preventing data leakage). A 80/20 sequential split without shuffling was enforced:
• **Training Set (train_data.csv):** 1,201 rows (Jan 20, 2021 – Aug 27, 2025)
• **Validation Set (validation_data.csv):** 301 rows (Aug 28, 2025 – Oct 22, 2026)

Phase 4: Model Training & Validation Performance

Seven algorithms plus a Naive Persistence Baseline were evaluated on the 301 held-out validation dates:

Model Name	Validation MAE (¢/L)	RMSE (¢/L)	MedAE (¢/L)	RMSLE	Status
Linear Regression	3.3834	4.8505	2.3955	0.0329	WINNER
Ridge Regression	3.4571	4.8582	2.4710	0.0329	Recommended Alt
Previous-Price Baseline	3.6578	4.9675	2.6000	0.0335	Persistence Benchmark
Decision Tree	3.8674	5.1668	2.9307	0.0350	Non-Linear Baseline

Model Name	Validation MAE (¢/L)	RMSE (¢/L)	MedAE (¢/L)	RMSLE	Status
Random Forest	3.8977	5.0981	3.0382	0.0347	Tree Ensemble
Gradient Boosting	3.9020	5.1002	3.1632	0.0344	Tree Ensemble
HistGradientBoosting	3.9133	5.0401	3.3383	0.0339	Fast Ensemble

Why Linear Regression Won: Linear Regression achieved the lowest MAE (3.38 ¢/L), outperforming the persistence baseline by 0.27 ¢/L. Tree ensembles overfit to training-period price boundaries (100–191 ¢/L) and partitioned feature space into step-functions, whereas linear regression successfully extrapolated smooth trend dynamics onto unseen future validation prices.

Phase 5: AI Governance & Deployment Application

• **Governance Protocols:** Scaler parameters locked to training bounds; automated retraining alerts when 30-day MAE > 3.50 ¢/L; mandatory analyst sign-off when predicted daily swings exceed 5.0 ¢/L.
• **Deployment Application (gas_price_app.py):** Operational Python app that computes all 14 lag features from 7 recent price inputs, applies Z-score scaling, and outputs next-day price predictions with 95% confidence intervals (± 1.96 * RMSE).

2. Studio Downloadable Artifacts Catalog

All generated datasets, trained model binaries, python application scripts, and visual charts are included in this ZIP package and accessible in the Studio panel:

File Name	Type / Format	Size	Description & Contents
cleaned_data.csv	CSV Dataset	25.8 KB	Cleaned daily unleaded fuel prices (1,516 rows, 2021-2026).
lag_features.csv	CSV Dataset	145.0 KB	14 engineered lag and rolling features + target (1,502 rows).
train_data.csv	CSV Dataset	354.1 KB	Standardized training partition (1,201 rows, 2021-2025).
validation_data.csv	CSV Dataset	86.4 KB	Standardized validation partition (301 rows, 2025-2026).
validation_predictions.csv	CSV Dataset	40.4 KB	Daily actual vs predicted prices across all 7 models.
validation_metrics.csv	CSV Summary	0.7 KB	Summary performance comparison table (MAE, RMSE, MedAE, RMSLE).
linear_regression.joblib	Model Binary	1.3 KB	Final trained Linear Regression model (Winning Model).
scaler.joblib	Scaler Binary	1.4 KB	Fitted StandardScaler instance for production inference.
all_trained_models.joblib	Model Bundle	1.1 MB	Complete dictionary containing all 7 fitted model instances.
gas_price_app.py	Python Script	5.0 KB	Full standalone gas price forecasting application script.
calgary_gas_prices_2021_2026.png	Chart PNG	231.0 KB	Historical price trend visualization highlighting peak and trough.

File Name	Type / Format	Size	Description & Contents
actual_vs_predicted_validation.png	Chart PNG	403.3 KB	Validation actual vs predicted time-series comparison plot.
calgary_gas_station_twilight.png	Visual Graphic	2.4 MB	Photorealistic graphic of a Calgary gas station at twilight.
energy_data_analytics_dashboard.png	Visual Graphic	2.0 MB	Photorealistic data science analytics dashboard visual.
fleet_fuel_budgeting_forecasting.png	Visual Graphic	2.1 MB	Photorealistic fleet fuel budgeting & forecasting visual.